COLLEGE OF ENGINEERING AND TECHNOLOGY
Mulavoor P.O, Muvattupuzha, Kerala – 686673

EduCredChain - Educational Credential Blockchain. A Hybrid Blockchain–PKI Architecture for Educational Credential Validation

MINI PROJECT REPORT

ABHIRAM S (ICE23CC003)
AMARNATH MOHAN (ICE23CC010)
MOHAMMED YASEEN (ICE23CC028)

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IN
COMPUTER SCIENCE AND ENGINEERING
(CYBER SECURITY)

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COLLEGE OF ENGINEERING AND TECHNOLOGY

Mulavoor P.O, Muvattupuzha, Kerala – 686673

DEPARTMENT OF

ARTIFICIAL INTELLIGENCE AND CYBER SECURITY

BONAFIDE CERTIFICATE

This is to certify that the project report entitled "**EduCredChain - Educational Credential Blockchain, A Hybrid Blockchain–PKI Architecture for Educational Credential Validation**" is a bonafide record of the project report presented by ABHIRAM S (ICE23CC003), AMARNATH MOHAN (ICE23CC010) and MOHAMMED YASEEN (ICE23CC028) during the academic year 2025 - 2026 towards the partial fulfilment of the requirement of the award of B. Tech Degree in Computer Science and Engineering (Cyber Security) of APJ Abdul Kalam Technological University, Thiruvananthapuram.

Prof. Haseena M K	**Prof. Ashna Yoosaf** **Prof. Asla M P** **Prof. Naveen K**	**Dr. Rosna P Haroon**
Assistant Professor Dept. of AD & CC	Assistant Professor Dept. of AD & CC	Professor & HOD Dept. of AD & CC
Project Guide	Project Coordinator	Head of the Department

DECLARATION

We Abhiram S, Amarnath Mohan and Mohammed Yaseen hereby declare that, this project report entitled *"EduCredChain - Educational Credential Blockchain, A Hybrid Blockchain–PKI Architecture for Educational Credential Validation*" is the bonafide work of mine carried out under the supervision of Prof. Haseena M K. We declare that to the best of our knowledge, the work report here in does not form part of any project report or dissertation the basis of which a degree or award was conferred on an earlier occasion on any other candidate the content of this report is not being presented by any other student to this or any other university for the award of degree.

Signature:
Name of student: ABHIRAM S
Uni.Register No: ICE23CC003

Signature:
Name of student: AMARNATH MOHAN
Uni.Register No: ICE23CC010

Signature:
Name of student: MOHAMMED YASEEN
Uni.Register No: ICE23CC028

Signature:
Name of the Guide: Prof. HASEENA M K

Signature:
Name of the Coordinator: Prof. ASLA M P

Signature:
Name of the Coordinator: Prof. ASHNA YOOSAF

Signature:
Name of the Coordinator: Prof. NAVEEN K

Countersigned with Name:
Dr. ROSNA P HAROON
HOD, Department of AI & CC
Ilahia College of Engineering and Technology

Date :
Place : Muvattupuzha

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Abhiram S

Date : Amarnath Mohan

Place : Muvattupuzha Mohammed Yaseen

ABSTRACT

EduCredChain is a decentralized and secure digital certificate management system designed to address the growing issue of certificate fraud and inefficient verification processes. Traditional methods of issuing and validating academic and professional credentials are often time-consuming, vulnerable to tampering, and reliant on centralized authorities. This project leverages blockchain technology to create a tamper-proof, transparent, and verifiable platform for issuing, storing, and validating certificates.

The system enables institutions to generate digital certificates that are cryptographically secured and stored on a blockchain network, ensuring immutability and authenticity. Each certificate is associated with a unique hash and can be verified instantly using QR codes or digital identifiers, eliminating the need for manual verification. Smart contracts automate the issuance and validation processes, reducing administrative overhead and enhancing trust among stakeholders.

EduCredChain improves data integrity, reduces fraud, and provides a scalable solution that can be integrated across multiple institutions and platforms. By combining blockchain with modern technologies, the project contributes to a more reliable, efficient, and secure credential verification ecosystem suitable for academic, corporate, and governmental use cases.

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LIST OF ABBREVIATIONS

	ABBREVIATIONS	DESCRIPTION
1.	PKI	Public Key Infrastructure
2.	SHA-256	Secure Hash Algorithm 256-bit
3.	RSA	Rivest–Shamir–Adleman (Public Key Cryptographic Algorithm)
4.	IPFS	InterPlanetary File System
5.	API	Application Programming Interface
6.	UI	User Interface
7.	DB	Database
8.	SQL	Structured Query Language
9.	SQLite	Self-contained SQL Database Engine
10.	HTTP	HyperText Transfer Protocol
11.	HTTPS	HyperText Transfer Protocol Secure
12.	TLS	Transport Layer Security
13.	SSL	Secure Sockets Layer
14.	DApp	Decentralized Application
15.	EVM	Ethereum Virtual Machine
16.	NFT	Non-Fungible Token
17.	PoW	Proof of Work
18.	PoS	Proof of Stake
19.	QR Code	Quick Response Code
20.	AI	Artificial Intelligence
21.	IoT	Internet of Things
22.	OWASP	Open Web Application Security Project

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CHAPTER-1
INTRODUCTION

Educational institutions increasingly rely on digital certificates to represent academic achievements, qualifications, and training credentials. However, traditional certificate management systems are vulnerable to issues such as forgery, unauthorized modification, and difficulty in verification. These limitations make it challenging for employers and institutions to confirm the authenticity of certificates quickly and reliably. As a result, there is a growing need for a secure, transparent, and efficient system that can ensure the integrity and trustworthiness of educational credentials.

The proposed project, *EduCredChain*, a Hybrid Blockchain–PKI architecture designed for secure educational credential verification. The system combines the immutability of blockchain technology with the security of cryptographic mechanisms such as SHA-256 hashing and public key encryption. When a certificate is issued, its data is securely processed and a cryptographic hash is generated. The certificate data is encrypted and stored in a database, while the hash is recorded on the blockchain. This approach ensures that any alteration to the certificate data can be easily detected, thereby preventing tampering and fraud.

In addition to certificate issuance, the system supports secure verification and revocation mechanisms. Users can verify a certificate by entering its unique identifier, after which the system checks data integrity and validates the stored hash against the blockchain. If necessary, certificates can also be revoked, and the revocation status is updated on the blockchain to maintain transparency. By integrating web technologies, cryptographic techniques, and blockchain infrastructure, the proposed system provides a reliable and tamper-resistant solution for managing and verifying educational credentials.

CHAPTER-2

LITERATURE SURVEY

1. **Verification and Validation of Certificate Using Blockchain**
   Authors: Krutant Dongare, Shivani Waghmare, Shruti Bobade, Sakshi Gogulwar, Achal Gothe, Vaishnavi Jayshingkar, Amol Dhankar
   [Nov 2025, IJRASET]

This paper proposes a blockchain-based system designed to provide a secure and transparent platform for issuing and verifying educational and professional certificates. Traditional certificate verification systems rely heavily on centralized databases and manual validation processes, which makes them vulnerable to forgery, duplication, and manipulation. The authors highlight the growing need for a reliable digital verification mechanism to ensure the authenticity of academic credentials and reduce fraudulent practices.

The proposed system uses blockchain technology to store cryptographic hashes of certificates, ensuring immutability and tamper resistance. To efficiently store certificate files, the system integrates the InterPlanetary File System (IPFS), where the actual certificate is stored while the blockchain maintains the reference hash. The system also provides an Android-based interface that enables institutions to issue certificates, students to manage them, and employers to verify authenticity instantly. This approach improves transparency, reduces verification time, and establishes a secure digital credential management system.

2. **Blockchain Technology for Secure Digital Certificate Generation and Verification**
   Authors: Kunal Godase, Omkar Ghate, Swapnil Kale, Saurabh Ghadage, Prof. Sudhakar Jadhav
   [MAR 2025, IJCRT]

This research focuses on developing a blockchain-based system to securely generate and verify digital academic certificates. The authors emphasize that traditional verification methods are inefficient, prone to forgery, and difficult to manage due to centralized storage. With the increasing number of graduates each year, institutions require a more reliable and scalable solution to verify credentials quickly and securely.

The proposed system uses Ethereum blockchain technology along with smart contracts to automate the certificate issuance and verification process. Certificate files are stored in the InterPlanetary File System (IPFS), while their cryptographic hash values and IPFS links are stored on the blockchain to maintain integrity. The system also incorporates technologies such as MetaMask for authentication, ReactJS for frontend development, and NodeJS for backend operations. By using decentralized technologies, the system ensures transparency, tamper-proof certificate storage, and instant verification without relying on centralized authorities.

3. Blockchain Driven Credential Issuing and Verification of Certificates
Authors: G. Raju, Lalitha Reddy Buchupalli, Aswini Thudimela, Kaveri Kodikondla, Kavitha Palaku, Divya Vadde
[APR 2025, GJEIIR]

This paper presents a blockchain-driven system for issuing and verifying educational certificates to address the growing problem of fake credentials and inefficient verification processes. The authors discuss how traditional methods require repeated submission and manual verification of certificates throughout a student's academic and professional journey. This process increases the risk of document loss, fraud, and administrative delays.

The proposed solution uses blockchain technology to create a decentralized and tamper-proof credential verification system. Certificates are issued and stored securely using cryptographic hashing and smart contracts, ensuring that once data is recorded on the blockchain it cannot be altered. The system also introduces a dual-chain architecture consisting of a certificate chain for original credentials and a correction chain for updates or modifications. This design enhances transparency, traceability, and efficiency, allowing students and organizations to verify certificates instantly without relying on intermediaries.

4. Certificate Verification and Validation Using Blockchain

Authors: Chittiprolu Saranya, Chittibomma Sai Sindhu, Ikkurthi Chandi Priya, Dantu Swathi, Anne Dhatri, Dr. A. Kalavathi

[APR 2025, JETNR]

This paper proposes a blockchain-based certificate verification system aimed at preventing the widespread issue of counterfeit academic certificates. In the traditional system, verification of educational credentials is often time-consuming and prone to manipulation due to the lack of transparency and secure storage mechanisms. The authors highlight that organizations and educational institutions require a reliable system to ensure the authenticity and integrity of academic records.

The proposed system converts traditional paper certificates into digital certificates and stores their cryptographic hash values on a blockchain network. Each certificate is assigned a unique digital signature and a QR code that can be scanned to verify its authenticity. When a certificate is scanned, the system retrieves the corresponding information from the blockchain and confirms whether the certificate is valid. By combining blockchain security with QR code verification, the system provides a fast, transparent, and tamper-proof solution for academic credential verification.

5. Blockchain-Based PKI:
Making Digital Certificate Management More Secure and Efficient

Authors: Sukhvinder Singh Bamber, Rajeev Kumar Dang, Naveen Dogra, Mohit Angurala

[AUG 2025, IJARSCT]

This paper discusses the role of digital certificates in ensuring secure communication and identity verification across digital platforms. Digital certificates form the core component of Public Key Infrastructure (PKI), enabling authentication, encryption, and secure data exchange across networks. The authors explain how traditional PKI systems rely heavily on centralized Certificate Authorities (CAs), which introduces vulnerabilities such as single points of failure, certificate authority compromise, and inefficient certificate revocation processes. The study also highlights the importance of SSL/TLS protocols in securing internet communications and explains how digital certificates help establish trust between communicating entities.

The paper proposes a blockchain-based PKI model to address the limitations of traditional certificate management systems. In this approach, blockchain acts as a decentralized ledger that securely records certificate issuance, verification, and revocation events. Smart contracts automate certificate lifecycle management, while distributed consensus ensures transparency and immutability of records. The proposed model improves security, scalability, and trust by eliminating reliance on centralized authorities. It also enables real-time certificate verification and reduces the risk of fraud, making it suitable for applications in e-governance, IoT environments, and secure digital identity systems.

6. **Blockchain-Based Decentralized Document Verification and Its Applications**
   Authors: Ayush Mishra, Sakshi Mehta, Bhavya Oza, Sumit Kumar, Hemant Kasturiwale
   [JAN 2025, JISEM]

This paper presents a blockchain-based decentralized framework for secure and efficient document verification, addressing the limitations of traditional methods such as high cost, manual effort, and vulnerability to fraud. The system integrates Ethereum blockchain and IPFS, where documents are stored off-chain in IPFS while only their cryptographic hash is recorded on the blockchain to ensure integrity and scalability. The workflow (shown in the system diagram on page 4) involves uploading documents through a decentralized application, generating a unique hash, and storing it immutably on the blockchain for verification.

The proposed system demonstrates significant improvements in performance and reliability, achieving high accuracy (99.9%), faster verification time, and reduced manual effort compared to traditional systems . It also ensures transparency, tamper-proof storage, and secure access control among stakeholders such as applicants, institutions, and employers. Overall, the research highlights the potential of blockchain technology to revolutionize document verification by providing a scalable, secure, and trustworthy solution, while also identifying challenges like scalability and integration with legacy systems.

CHAPTER-3

EXISTING SYSTEM

3.1 Introduction

Several digital credential management systems have been developed to enable secure issuance and verification of certificates. Among them, Blockcerts, OpenCerts, and DigiLocker are widely recognized models that demonstrate different approaches to handling digital credentials.

Blockcerts is an open standard for issuing and verifying digital certificates using blockchain technology. In this model, certificate data is digitally signed and its hash is stored on the blockchain, while the actual certificate is shared with the user. Verification is performed by matching the certificate's hash with the blockchain record. OpenCerts is a blockchain-based platform that enables institutions to issue certificates as digitally signed files with embedded metadata. These certificates can be shared directly by users and verified by validating the digital signature and corresponding blockchain entry. The system focuses on standardized digital credential formats and verification. DigiLocker is a centralized digital document storage system developed by the Government of India for storing and accessing official records. Institutions can issue certificates directly to users' accounts, where they are stored securely. Verification is carried out by retrieving documents from the centralized system and validating them against issuing authorities.

These systems represent different architectural approaches to digital credential management, ranging from traditional centralized systems to modern blockchain-based decentralized models. Traditional systems primarily rely on physical documents or centralized databases maintained by institutions, where verification is performed manually or through direct authority validation. In contrast, newer approaches such as Blockcerts and OpenCerts utilize decentralized blockchain networks for secure and tamper-evident verification, while platforms like DigiLocker adopt a centralized digital repository for streamlined access and management. Together, these models highlight the evolution of credential management from manual and institution-dependent processes to more secure, digital, and technology-driven solutions.

3.2 Advantages

1. Digital Accessibility, Certificates can be accessed, stored, and shared electronically without physical handling.
2. Faster Verification, Digital systems reduce the time required to verify credentials compared to manual methods.
3. Reduced Paperwork, Minimizes reliance on physical documents, simplifying storage and management.
4. Improved Record Management, Structured storage helps institutions organize and retrieve data efficiently.
5. Easy Integration, Systems can be integrated with existing institutional or government platforms for streamlined operations.

3.3 Disadvantages

1. Centralized Dependency, Many systems rely on a single authority, creating a potential single point of failure.
2. Limited Transparency, Verification processes may not always be openly visible or independently verifiable.
3. Security Risks, Centralized databases are vulnerable to data breaches, tampering, or unauthorized access.
4. Scalability Issues, Managing large volumes of certificates across multiple institutions can be challenging.
5. Interoperability Gaps, Lack of standardization makes it difficult for different systems to communicate seamlessly.

CHAPTER-4
PROPOSED SYSTEM

4.1 Introduction

The proposed system, EduCredChain, is a hybrid Blockchain–PKI based architecture designed to provide a secure, transparent, and reliable platform for managing educational credentials. Traditional certificate management systems often depend on manual verification processes or centralized databases, which are prone to data tampering, certificate forgery, unauthorized access, and inefficiencies in validation. To overcome these limitations, the proposed system integrates blockchain technology with robust cryptographic mechanisms to ensure the authenticity, integrity, and confidentiality of certificate data. In this system, when a certificate is issued, its data is derived to a unique cryptographic hash using SHA-256 algorithm. This hash acts as a digital fingerprint of the certificate, ensuring that even the slightest change in data results in a completely different hash value. The sensitive certificate data is encrypted using the issuer's private key, ensuring secure storage and controlled access, and is then stored in a structured database. Simultaneously, the generated hash is recorded on the blockchain, creating an immutable and tamper-proof reference. Since blockchain records are inherently immutable and distributed, any attempt to modify the certificate data in the database can be easily detected. During verification, the system retrieves the stored certificate data, regenerates its hash, and compares it with the hash stored on the blockchain. If both hashes match, the certificate is considered authentic; otherwise, it is flagged as tampered or invalid. The system also incorporates certificate verification and revocation mechanisms to support the complete lifecycle of credentials. Users can verify certificates through a web-based interface by entering a unique certificate ID, enabling quick and reliable validation. In cases where a certificate needs to be invalidated, authorized users can revoke it, and the revocation status is permanently recorded on the blockchain. This ensures transparency, prevents misuse of revoked certificates, and enhances trust among institutions, employers, and users. Overall, EduCredChain provides a scalable, secure, and tamper-resistant solution for modern digital credential management.

4.2 Advantages

1. Enhanced Security: Integration of cryptographic hashing and encryption ensures protection against certificate tampering and unauthorized access.
2. Tamper Resistance: Storing certificate hashes on the blockchain ensures data immutability and prevents fraudulent modifications.
3. Reliable Verification: Certificates can be verified quickly and accurately through blockchain validation.
4. Transparency: Blockchain provides a transparent and traceable record of certificate issuance and revocation.
5. Decentralized Trust: Verification does not rely solely on a central authority, increasing system reliability.
6. Efficient Credential Management: Automates certificate issuance, verification, and revocation processes.
7. Scalability: The architecture can be extended to support multiple institutions and credential types.

4.3 Limitations

1. Implementation Complexity: Integrating blockchain, cryptography, and web systems requires technical expertise.
2. Higher Initial Setup Cost: Blockchain network setup and maintenance may require additional resources.
3. Transaction Latency: Blockchain operations may introduce slight delays compared to traditional databases.
4. Dependency on Cryptographic Key Management: Loss or compromise of cryptographic keys may affect data accessibility.
5. System Integration Challenges: Integration with existing institutional systems may require additional customization.

CHAPTER-5

SYSTEM REQUIREMENTS

5.1 Hardware Requirements

The proposed system does not require specialized hardware and can be implemented using standard computing resources. The hardware requirements include:

Processor: Intel Core i3 or higher
RAM: Minimum 4 GB (8 GB recommended)
Storage: Minimum 20 GB free disk space
Network: Stable internet connection for blockchain communication
Display: Standard monitor with minimum 1366×768 resolution

5.2 Software Requirements

The system is developed using modern programming and web technologies to ensure security and efficiency.

Operating System: Windows / Linux / macOS
Programming Language: Python 3.x
Web Technologies: HTML, CSS, JavaScript
Database: SQLite
Cryptography Libraries: Python Cryptography Library (RSA, SHA-256)
Blockchain Platform: Local or test blockchain network
Development Environment: VS Code / PyCharm
Web Browser: Google Chrome / Microsoft Edge / Firefox

5.3 Expected System Requirements

The expected system requirements define the core functionalities and operational behavior of the EduCredChain system. These requirements outline how the system performs certificate issuance, verification, and secure data management using blockchain and cryptographic techniques.

REQ 1: Certificate Issuance

Description: The system must allow authorized users to issue digital certificates through a web interface.

Input: Certificate details such as name, course ID, certificate number, issue date, and grade.

Output: Encrypted certificate data stored in the database and corresponding hash recorded on the blockchain

REQ 2: Certificate Hash Generation

Description: The system must generate a unique SHA-256 hash for each certificate to ensure integrity.

Input: Normalized certificate data fields.

Output: A fixed-length cryptographic hash representing the certificate data.

REQ 3: Secure Data Storage

Description: The system must securely store certificate data using encryption in an off-chain database.

Input: Certificate details from the issuance module.

Output: Encrypted records stored in the SQL database with associated certificate ID.

REQ 4: Blockchain Integration

Description: The system must record certificate hashes on a blockchain network for immutability.

Input: Generated certificate hash.

Output: Transaction recorded on the blockchain with hash and identifier.

REQ 5: Certificate Verification

Description: The system must allow users to verify certificates using a unique certificate ID.

Input: Certificate ID provided by the user.

Output: Verification result indicating whether the certificate is valid, tampered, or invalid.

REQ 6: Certificate Revocation

Description: The system must enable authorized users to revoke certificates when required.

Input: Certificate ID selected for revocation.

Output: Updated revocation status stored in the database and recorded on the blockchain.

REQ 7: User Interface Interaction

Description: The system must provide a user-friendly web interface for certificate operations.

Input: User actions such as issuing, verifying, or revoking certificates.

Output: Display of results, confirmations, and status messages.

REQ 8: Data Integrity Validation

Description: The system must validate certificate integrity by comparing recalculated hashes with blockchain records.

Input: Stored certificate data and blockchain hash.

Output: Confirmation of data integrity or detection of tampering.

REQ 9: Blockchain Transaction Monitoring

Description: The system must log and track blockchain transactions related to certificate operations.

Input: Blockchain transaction details.

Output: Transaction status and logs for transparency and auditing.

5.4 Feasibility Study

A feasibility study evaluates whether the proposed system is practical and beneficial for implementation.

1. Technical Feasibility

The proposed system is technically feasible as it utilizes widely available technologies such as Python, web development frameworks, cryptographic libraries, and blockchain platforms. The required tools and libraries are open-source and well-documented, making development and implementation achievable with available resources.

2. Operational Feasibility

The system is designed with a simple and intuitive web interface, allowing issuers and users to interact with the system easily. Institutions can issue certificates digitally, and users can verify them quickly using the certificate ID. This improves operational efficiency and reduces manual verification processes.

3. Economic Feasibility

The system is economically feasible because it primarily uses open-source technologies and does not require expensive infrastructure. Implementation costs are minimal, limited mainly to development time and basic computing resources.

4. Legal Feasibility

The proposed system complies with general digital security and data protection principles. It ensures the confidentiality and integrity of certificate data using encryption and cryptographic techniques. Additionally, blockchain provides a transparent and verifiable record of credential issuance and revocation.

5. Schedule Feasibility

The development of the proposed system can be completed within a reasonable timeframe using available technologies. Since the system components such as web interfaces, database management, and cryptographic libraries are readily available, the project can be implemented within the planned academic project schedule.

5.5 Work Plan

The development of the EduCredChain: A Hybrid Blockchain–PKI Architecture for Educational Credential Verification was planned and executed in a systematic manner. The project was divided into several phases to ensure smooth progress and timely completion. The following work plan outlines the major tasks carried out during the development process.

1. Requirement Analysis & Feasibility Study

   Gather functional and non-functional requirements. Analyze system feasibility (technical, operational, economic, legal). Identify required technologies and tools
   Duration: 1 Week

2. System Design & Architecture Planning

   Design overall system architecture and module structure. Plan database schema for certificate storage. Design blockchain interaction workflow. Prepare UML diagrams and system flow diagrams
   Duration: 1 Week

3. Backend Development

   Develop backend logic using Python. Implement certificate issuance, verification, and revocation modules. Integrate cryptographic functions such as SHA-256 hashing and RSA encryption. Implement API communication between frontend and backend
   Duration: 2 Weeks

4. Database and Blockchain Integration

   Implement SQLite database for certificate data storage. Store encrypted certificate
   data and hash values in the database. Integrate blockchain network for storing certificate hashes. Implement blockchain-based verification and revocation mechanisms
   Duration: 1 Week

5. Web Interface Development

   Design and implement user interface using HTML, CSS, and JavaScript. Develop pages for certificate issuance, verification, and revocation. Ensure smooth interaction between frontend and backend through APIs
   Duration: 1 Week

6. System Integration and Testing

   Integrate all system modules including frontend, backend, database, and blockchain. Perform functionality testing and debugging. Verify the accuracy of certificate verification and revocation processes. Conduct security and integrity validation

   Duration: 1 Week

7. Documentation and Final Deployment

   Prepare project documentation and technical report. Create system diagrams, flowcharts, and explanations. Deploy the system for demonstration and presentation

   Duration: 1 Week

CHAPTER-6

SYSTEM MODEL

6.1 ARCHITECTURE DESIGN

Architecture Design refers to the high-level structure of a software system, defining its components, modules, and their interactions. The architecture of EduCredChain follows a hybrid model integrating a web-based user interface, backend processing system, database, and blockchain network. Users interact through the interface to issue, verify, or revoke certificates, while the backend handles data processing, hashing, and encryption. Certificate data is securely stored in the database, and its cryptographic hash is recorded on the blockchain to ensure immutability and integrity. This layered architecture enables secure communication between components and provides a reliable, tamper-resistant credential management system.

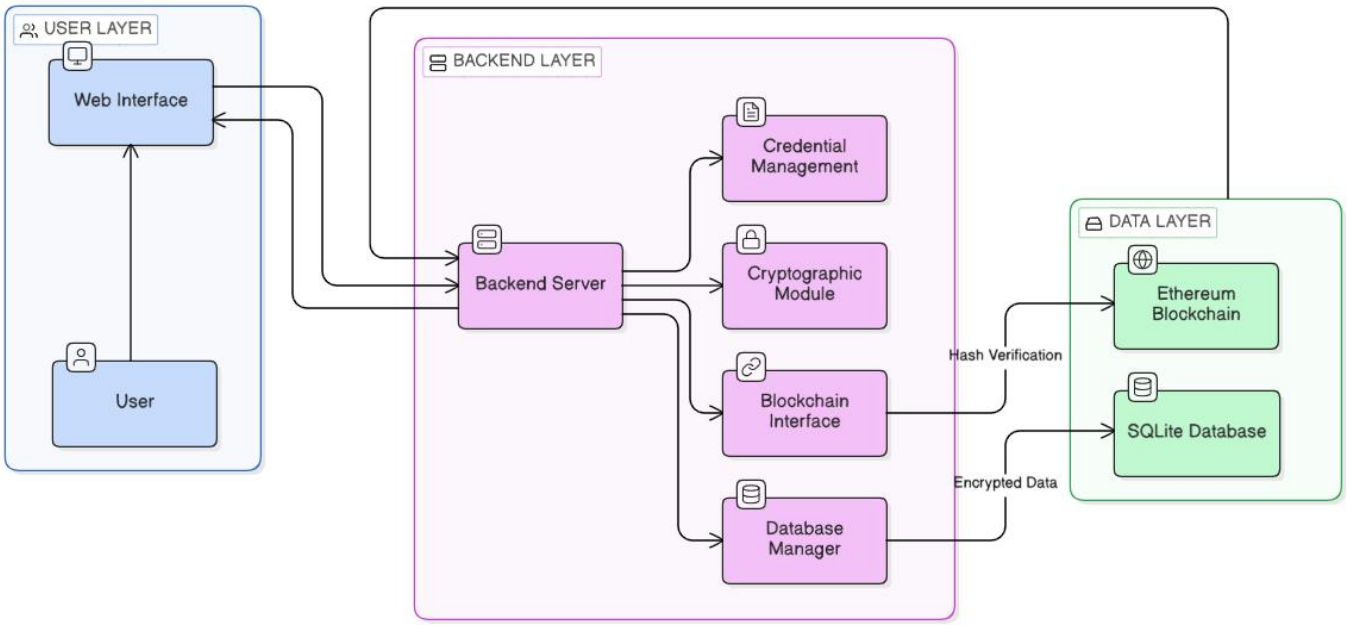

Fig 5.1 Architecture Diagram

6.2 USE CASE DIAGRAM

A Use Case Diagram is a visual representation of the interactions between users (actors) and the system. It shows the functional requirements of the system by depicting the various use cases (functions) that users can perform and how they interact with different components.

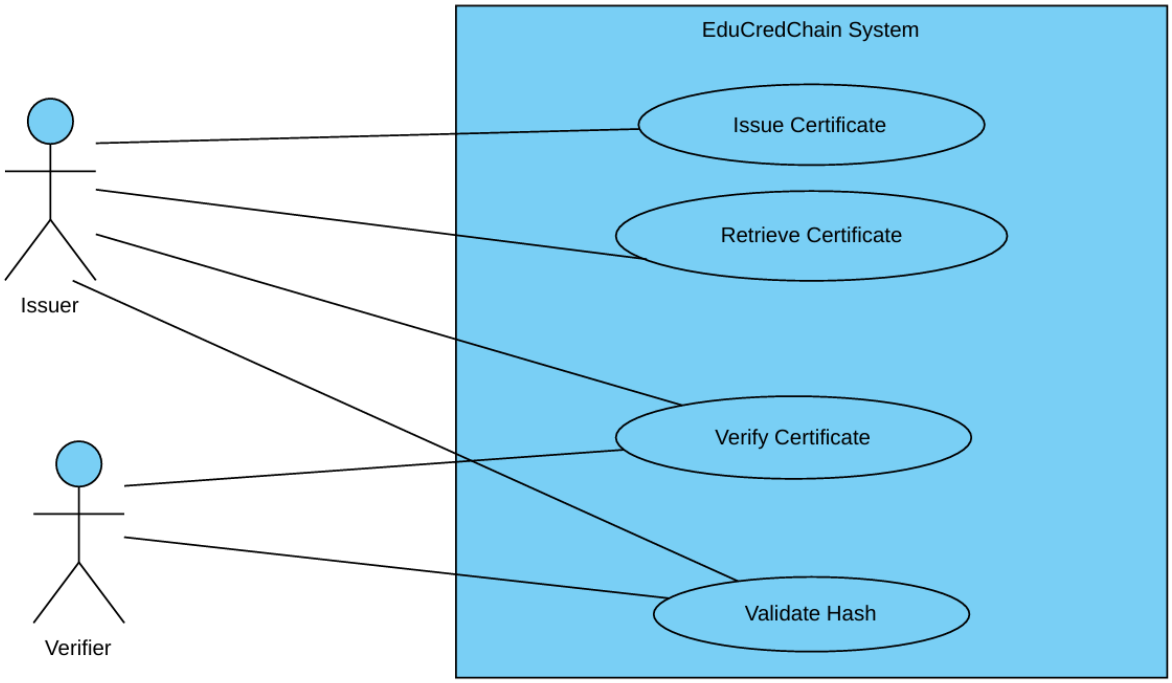

Fig 5.2 Use case diagram

6.3 SEQUENCE DIAGRAM

The sequence diagram shows interactions between the user, backend, database, and blockchain. The system processes certificate data, generates a hash, and stores it securely while recording the hash on the blockchain. During verification, the hash is recalculated and compared with the blockchain record to confirm authenticity.

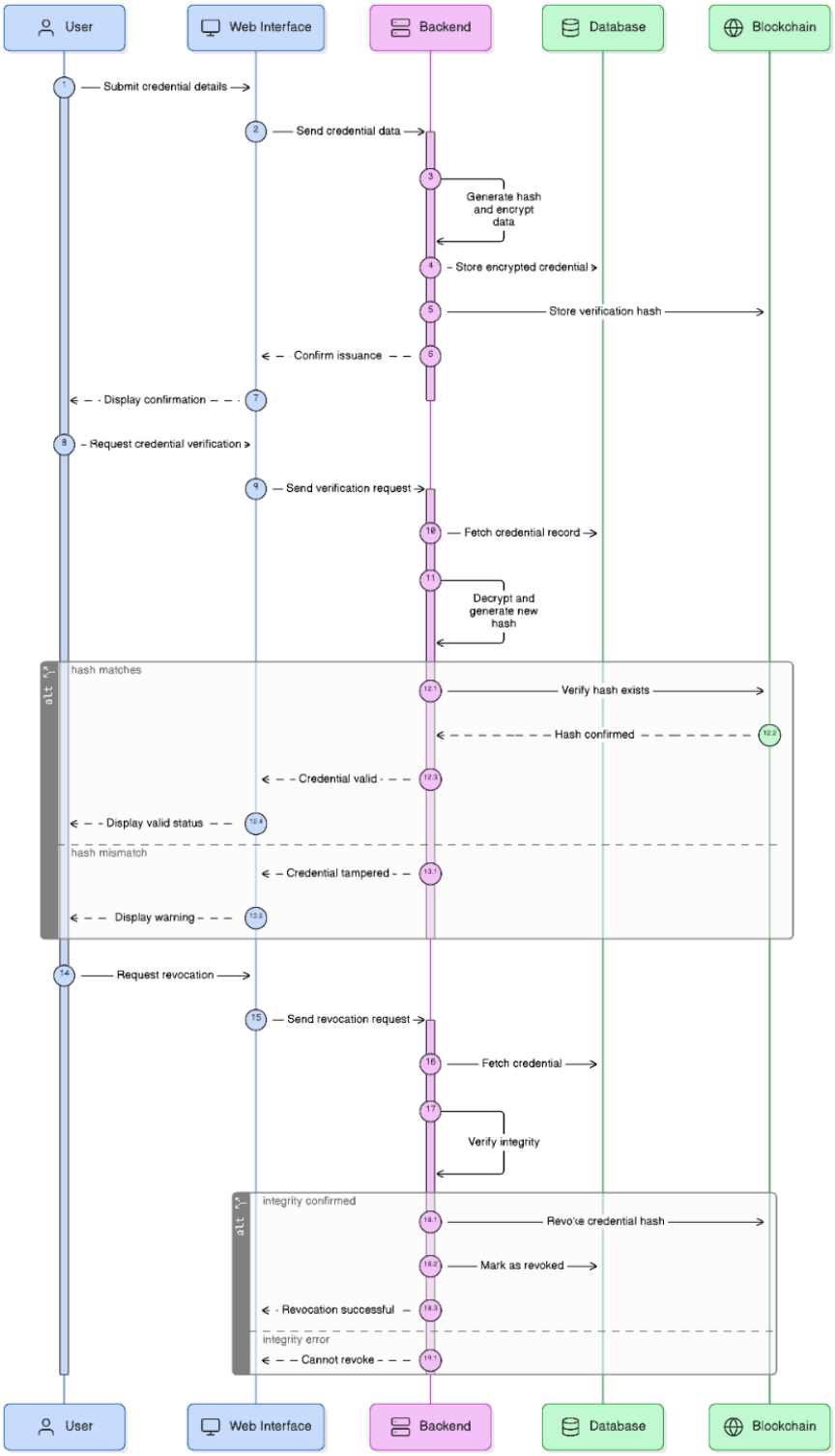

Fig 5.3 Sequence Diagram

6.4 ACTIVITY DIAGRAM

An Activity Diagram is a flowchart that models the workflow of a system or process. It shows the sequence of activities, decisions, and parallel processes, helping to visualize the logic of complex operations and the flow of control from one activity to another.

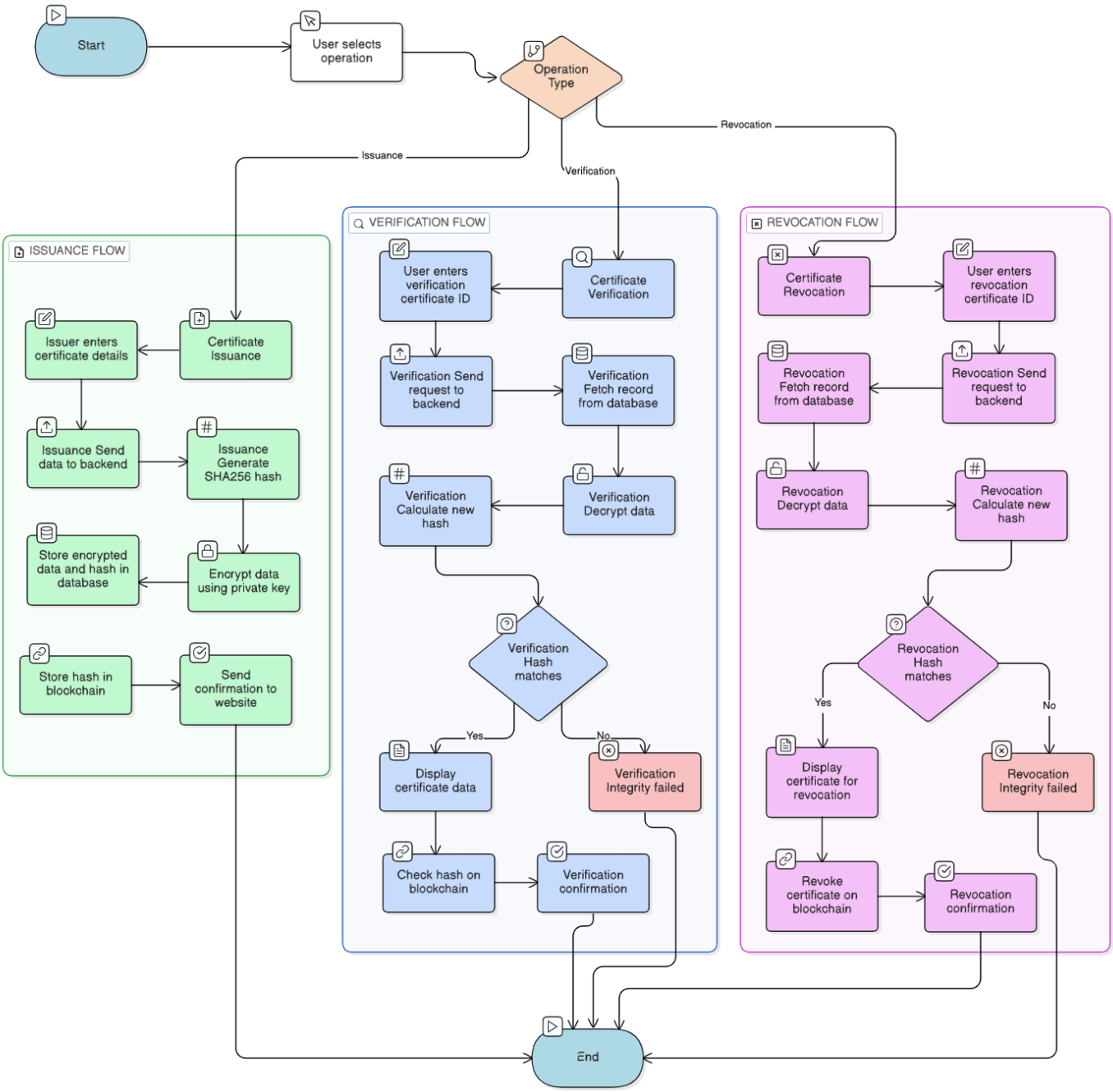

Fig 5.4 Activity Diagram

6.5 ENTITY-RELATIONSHIP DIAGRAM

The ER diagram of the EduCredChain system represents the relationships between key entities such as certificates, users (issuers/verifiers), and blockchain records. It illustrates how certificate details are stored in the database along with associated attributes like certificate ID, student information, course data, and issue date. The diagram also shows the linkage between certificate records and their corresponding hash values stored on the blockchain. This structured representation helps in understanding data organization, relationships, and efficient management of certificate information within the system.

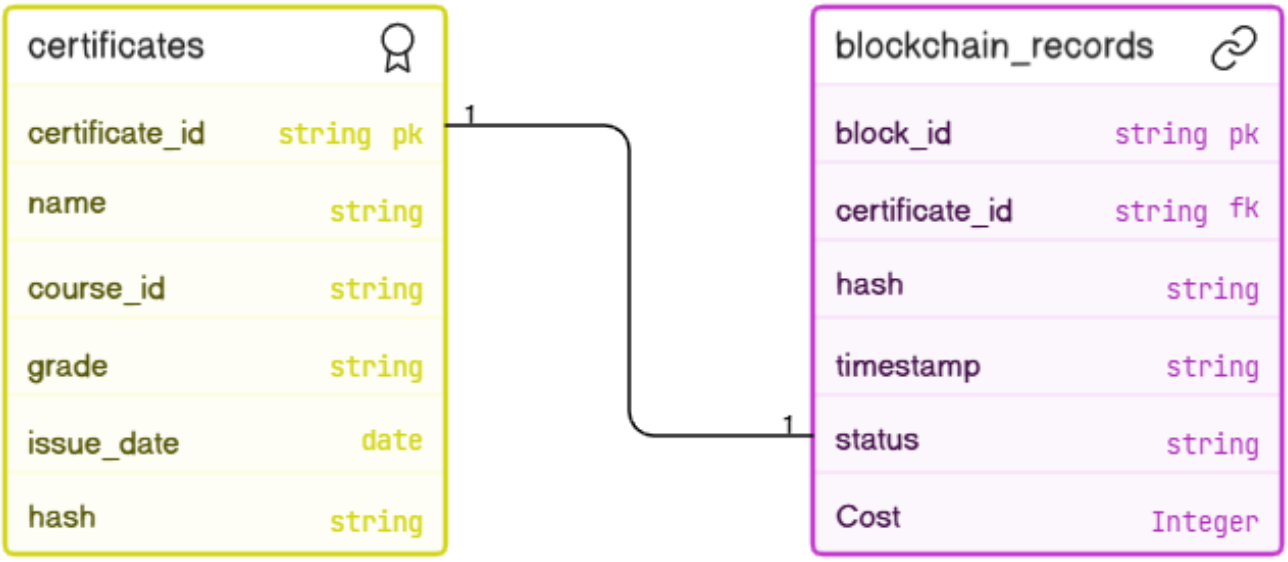

Fig 5.5 Entity-Relationship Diagram

CHAPTER-7
IMPLEMENTATION

The system is implemented using a combination of web technologies, Python-based backend processing, cryptographic techniques, and a blockchain network to ensure secure and efficient certificate management. This chapter describes the key components of the implementation, including the certificate processing module, secure data storage, and blockchain-based verification and revocation mechanisms.

7.1 Blockchain Infrastructure Module

7.1.1 Core Blockchain Framework

The Blockchain Infrastructure Module acts as the core foundation, enabling secure, decentralized, and tamper-proof certificate management.Blockchain technology is used to store cryptographic hashes of issued certificates, ensuring immutability. Once recorded on the blockchain, certificate data cannot be altered or deleted.

7.1.2 Distributed Ledger Implementation

Ethereum (locally implemented via Ganache) is used to maintain a distributed ledger. The ledger is accessible to authorized users, ensuring transparency and trust. This setup enables a controlled blockchain environment for development and testing.

7.1.3 Transaction Validation Mechanism

Certificate-related processes such as issuance and verification are validated through blockchain mechanisms. transaction is recorded securely, maintaining consistency across the network. This eliminates the need for centralized control in verification processes.

7.1.4 Security and Integrity Features

The infrastructure enhances data integrity by ensuring tamper-proof storage of certificate hashes. It enables real-time verification of credentials using blockchain records. The system remains reliable, scalable, and resistant to fraud due to its decentralized design.

7.2 Certificate Hash Generation

7.2.1 Input Normalization

Certificate fields are preprocessed by converting text values to lowercase and removing unnecessary spaces. The name, Course ID, and certificate number are normalized, while issue date and grade are preserved in original format. This ensures consistency and prevents mismatches during verification.

7.2.2 Hash Generation

The concatenated data is passed into the SHA-256 algorithm. A fixed-length, tamper-proof hash is generated for each certificate.

7.2.3 Mathematical Representation

The hash generation process is defined as:

$$\mathbf{H = SHA256(lower(n) \,||\, lower(c) \,||\, lower(cn) \,||\, d \,||\, g)}$$

Where:

H = Generated certificate hash
n = Name
c = Course ID
cn = Certificate Number
d = Issue Date
g = Grade
|| = Concatenation operator

7.3 Data Storage Architecture

7.3.1 Hybrid Storage Model

The system uses a hybrid architecture dividing data between off-chain and on-chain storage. This approach ensures efficiency, scalability, and security in managing certificates.

7.3.2 On-Chain Storage Layer

Only cryptographic hashes of certificates are stored on the blockchain (Ethereum). Minimal identifiers are stored along with the hash to ensure immutability and tamper resistance. This reduces storage cost and avoids inefficiency of storing full certificates on-chain.

7.3.3 Off-Chain Storage Layer

Actual certificate data and metadata are encrypted and stored in a SQL-based database. Stored details include student information, course data, certificate number, issue date, and grade. This enables faster access and reduces storage overhead.

7.3.4 Certificate Issuance Process

Certificate data is first stored securely in the database. A hash is generated from the stored data. The generated hash is recorded on the blockchain using smart contracts.

7.3.5 Verification Process

The system retrieves certificate data from the database during verification. A new hash is generated and compared with the blockchain record. Matching hashes confirm the authenticity of the certificate.

7.3.6 System Benefits

Ensures data integrity through hash validation. Reduces blockchain load and operational costs. Provides high performance and scalability for large-scale adoption.

CHAPTER-8
RESULT & DISCUSSION

8.1 Index Page

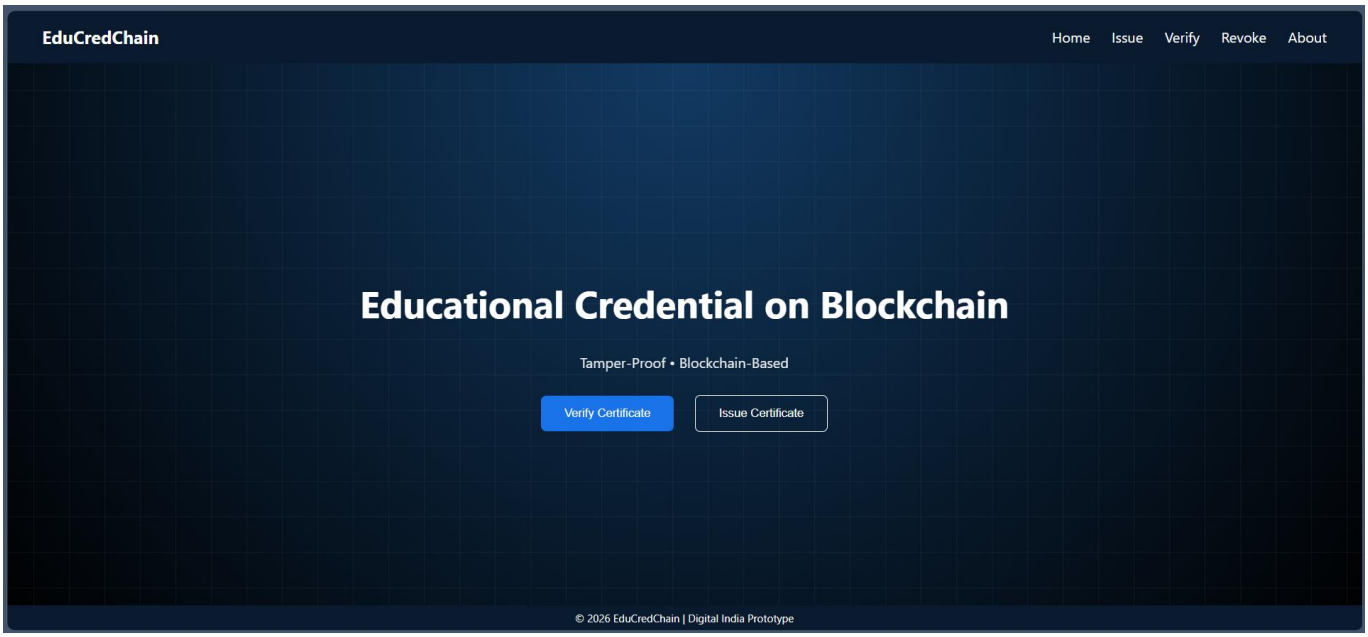

fig 8.1 Index Page

Figure 8.1 illustrates the index page of the EduCredChain system, which serves as the main interface for users to access different functionalities. It provides navigation options for certificate issuance, verification, and revocation. The interface is designed to be simple and user-friendly, allowing users to interact with the system efficiently. This page acts as the entry point for all certificate management operations.

8.2 Certificate Issue Page

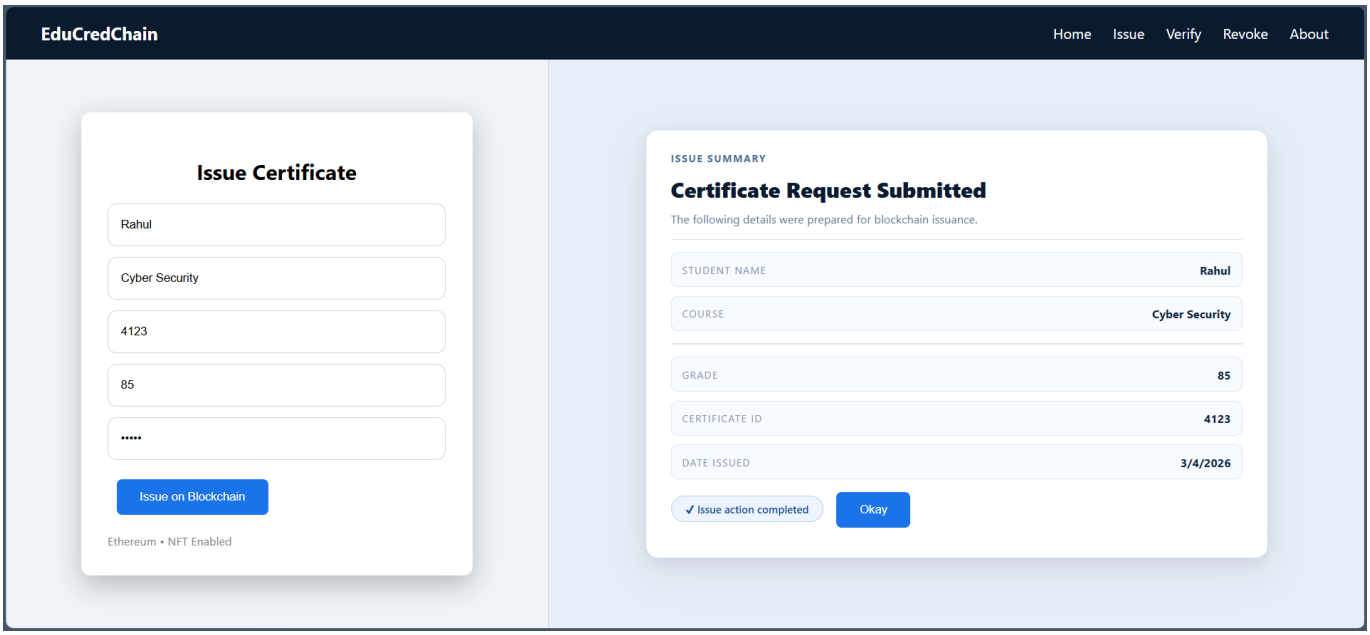

fig 8.2 Certificate Issue

Figure 8.2 illustrates the certificate issuance interface where authorized users input essential certificate details such as name, course ID, certificate number, issue date, and grade through a structured web form. Once the data is submitted, the backend system preprocesses and normalizes the inputs before generating a unique cryptographic hash using the SHA-256 algorithm. The certificate data is then securely encrypted and stored in the database, ensuring confidentiality and efficient retrieval. Simultaneously, the generated hash is recorded on the blockchain using smart contracts, providing an immutable reference for future verification. This interface ensures a secure, consistent, and tamper-resistant process for issuing digital certificates while maintaining data integrity and transparency.

8.3 Certificate Verification Page

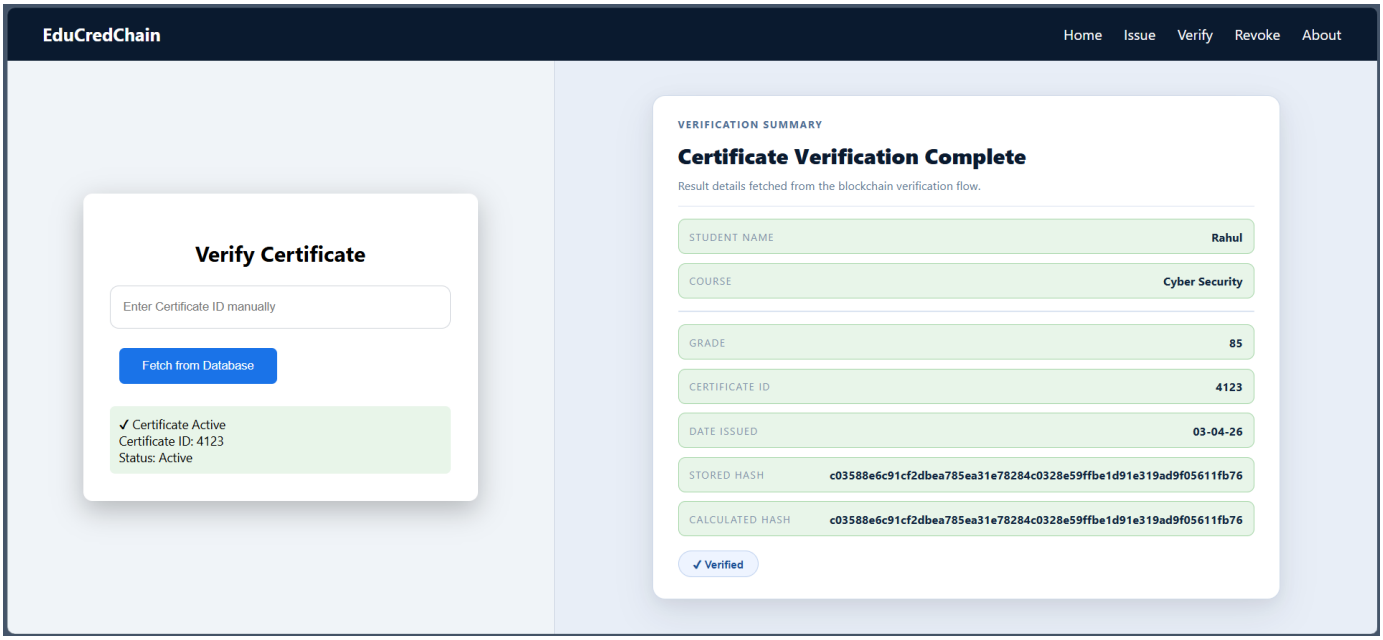

fig 8.3 Certificate Verification

Figure 8.3 illustrates the certificate verification interface that allows users to validate the authenticity of a certificate using its unique identifier. When a request is submitted, the system retrieves the corresponding certificate data from the database and processes it through the backend. A new cryptographic hash is generated using the same method applied during issuance, ensuring consistency in verification. This recalculated hash is then compared with the hash stored on the blockchain to check for any discrepancies. Based on the comparison, the system determines whether the certificate is valid, tampered, or invalid, providing a reliable and secure verification mechanism.

8.4 Certificate Revocation

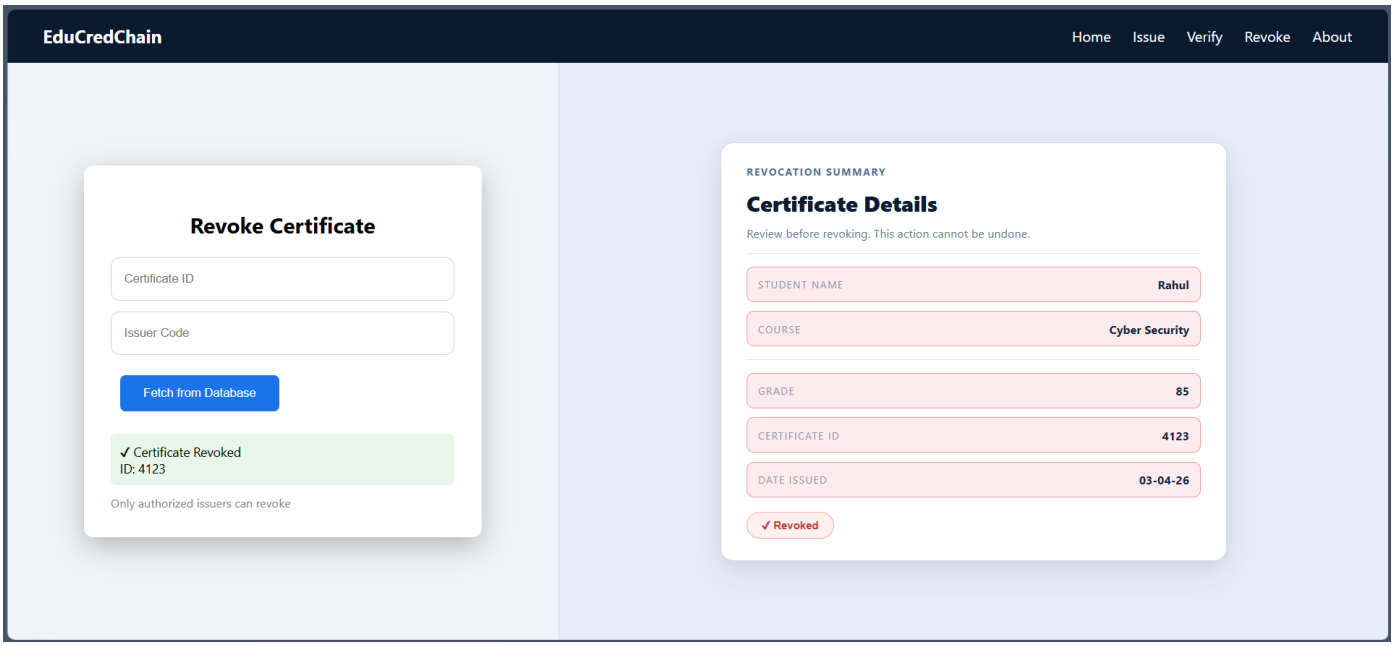

fig 8.4 Certificate Revocation

Figure 8.4 illustrates the certificate revocation interface used by authorized users to invalidate a certificate when required. Through this interface, users can specify the certificate to be revoked using its unique identifier, after which the backend processes the request securely. The system updates the certificate status in the database and records the revocation status on the blockchain to ensure immutability and transparency. During future verification attempts, the system checks this status to prevent the use of revoked certificates. This mechanism strengthens the overall integrity and trustworthiness of the credential management system.

8.5 Retrieving Tampered Data from Database

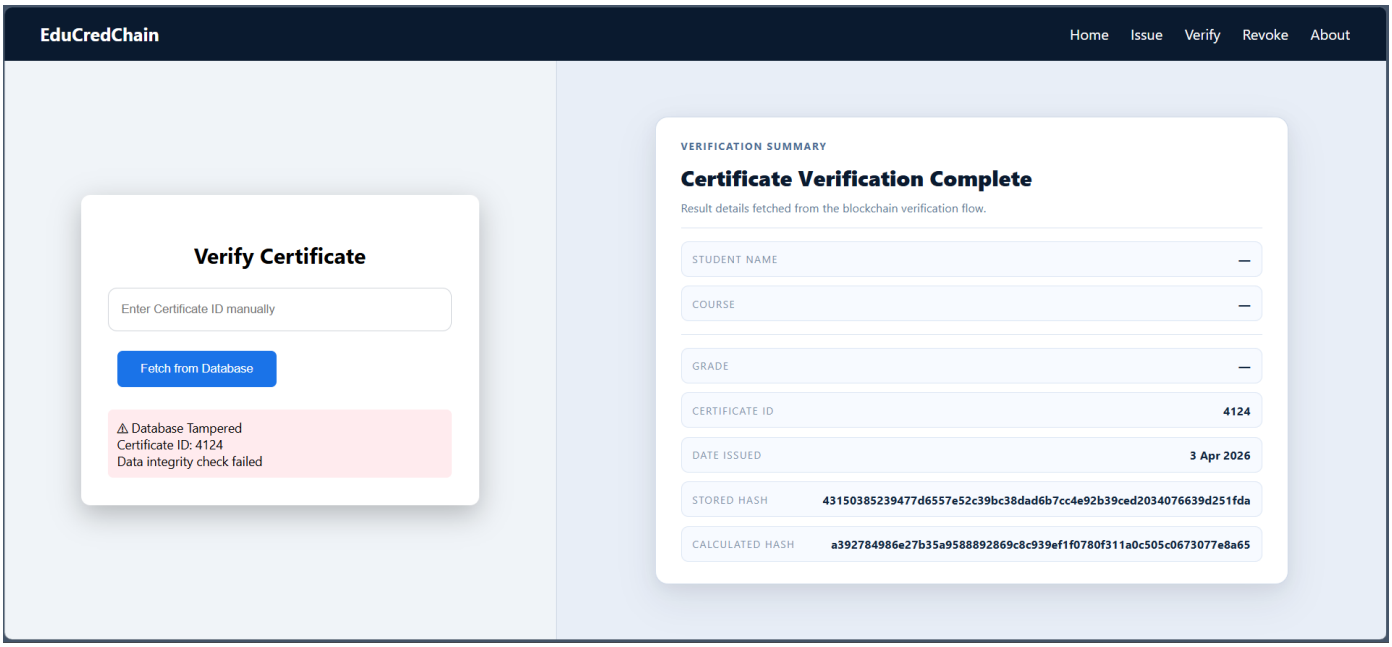

fig 8.5 Retrieving Tampered Data from Database

Figure 8.5 illustrates the process of retrieving tampered certificate data from the database to demonstrate the system's integrity validation mechanism. When certificate data is accessed, the system recalculates its cryptographic hash using the stored values. This newly generated hash is then compared with the original hash recorded on the blockchain. If any mismatch is detected, it indicates that the certificate data has been altered or compromised. This process highlights the system's ability to identify tampering and effectively prevent fraudulent use of modified certificates.

8.6 Checking an invalid Certificate

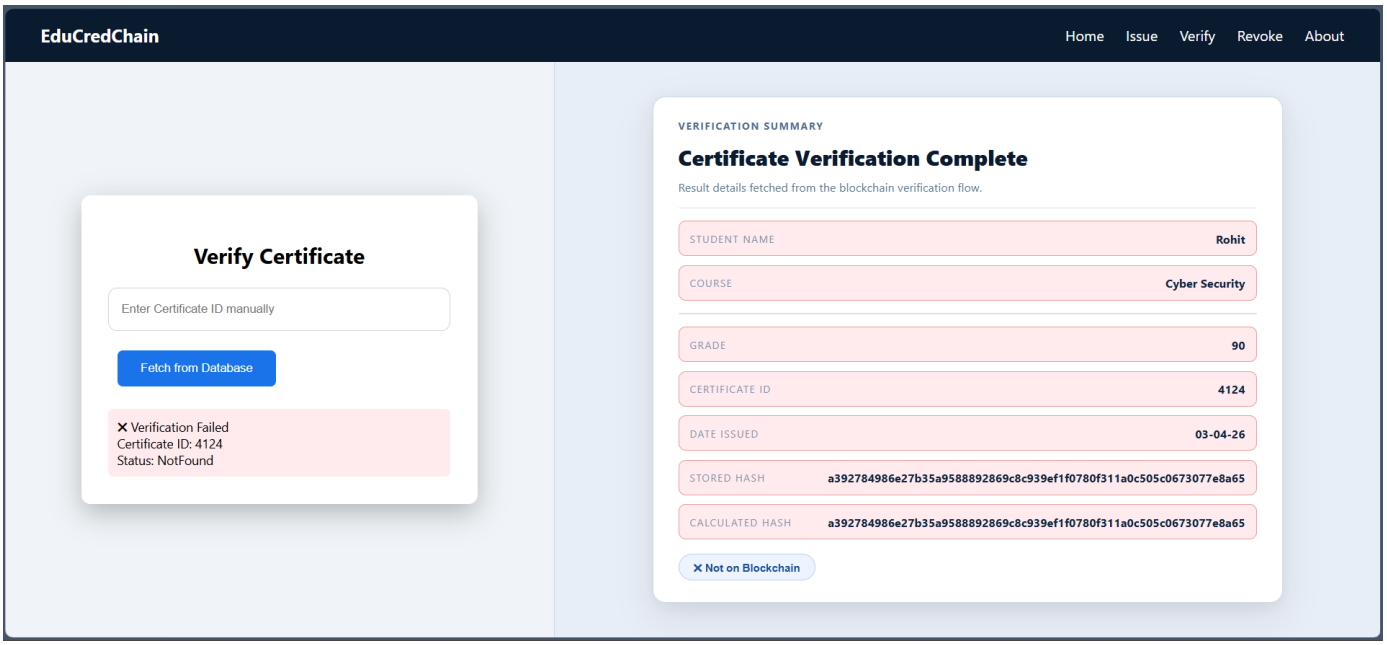

fig 8.6 Checking an invalid Certificate

Figure 8.6 illustrates the validation of an invalid certificate within the system, where a certificate that does not correspond to any stored or recognized record is submitted for verification. The system attempts to retrieve the associated data and generate a hash, but fails to find a matching blockchain entry. As a result, the certificate is identified as invalid and rejected by the system. This process ensures that only legitimate and registered certificates are accepted. It enhances the reliability and robustness of the verification mechanism by preventing unauthorized or fake credentials.

8.7 Checking a Revoked Certificate

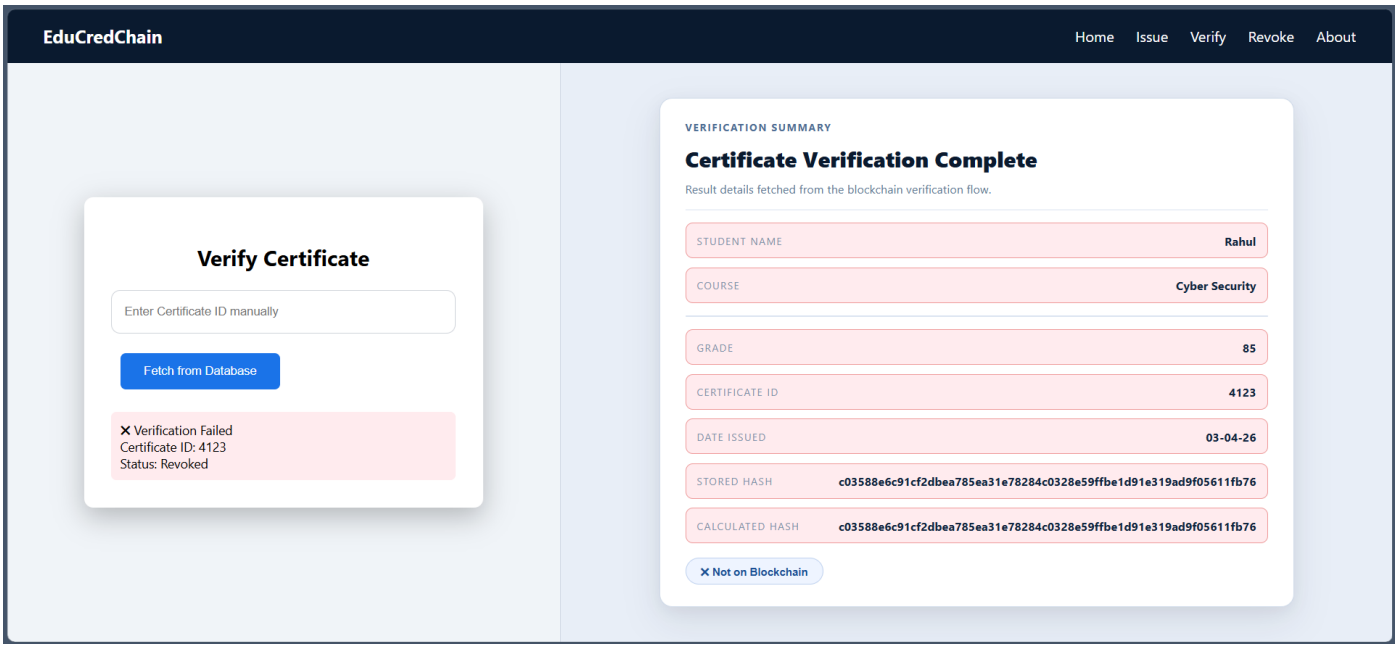

fig 8.7 Checking a Revoked Certificate

Figure 8.7 illustrates the verification of a revoked certificate in the system, where a certificate with valid data is checked against its current status. During the verification process, the system retrieves the certificate details and compares the generated hash with the blockchain record to confirm integrity. In addition to hash validation, the system checks the revocation status stored on the blockchain. If the certificate is marked as revoked, it is flagged as invalid despite having correct data. This ensures that revoked credentials cannot be reused or misused, maintaining the integrity of the system.

CHAPTER-8

FUTURE SCOPE

EduCredChain system lies in expanding its capabilities to support a wider range of institutions and real-world applications. The system can be scaled to integrate multiple universities, certification authorities, and organizations into a unified blockchain network. Advanced technologies such as smart contracts can be incorporated to automate certificate issuance, verification, and revocation processes. Additionally, mobile applications and QR code-based verification can be developed to enhance accessibility and user convenience. The adoption of scalable blockchain solutions and cloud deployment can further improve system performance and reliability.

1. Multi-Institution Integration

The system can be extended to support multiple educational institutions and certification bodies on a shared platform. This allows different organizations to issue and verify certificates within a unified blockchain network. It enhances interoperability and reduces duplication of verification efforts. Such integration increases trust and collaboration between institutions. It also enables a standardized approach to digital credential management.

2. Smart Contract Automation

Smart contracts can be implemented to automate certificate issuance, verification, and revocation processes. These contracts execute predefined rules without human intervention, ensuring transparency and consistency. Automation reduces manual errors and speeds up operations. It also enhances security by eliminating reliance on centralized control. This makes the system more efficient and reliable.

3. Mobile Application Development

Developing a mobile application can improve accessibility and user convenience. Users can issue, verify, and manage certificates directly from their smartphones. Mobile apps can provide real-time notifications and quick verification features. This enhances usability for both institutions and end-users. It also increases adoption by making the system more user-friendly.

4. Integration with Government Systems

The system can be integrated with government databases and digital identity platforms. This ensures official validation and increases the credibility of issued certificates. It can help in streamlining national-level education and employment verification processes. Such integration reduces fraudulent claims and enhances trust. It also supports wider adoption across public sectors.

5. Enhanced Security Mechanisms

Advanced security techniques can be incorporated to strengthen the system. Features like multi-signature authentication and zero-knowledge proofs can be implemented. These methods provide additional layers of protection for sensitive data. Proper key management practices can further improve security. This ensures that the system remains robust against cyber threats.

6. Scalable Blockchain Solution

To handle a large number of transactions, scalable blockchain solutions can be adopted. Technologies like Layer-2 scaling or efficient consensus mechanisms can improve performance. This reduces transaction latency and operational costs. It ensures that the system can support growth in users and institutions. Scalability is essential for real-world deployment.

7. QR Code-Based Verification

QR codes can be added to certificates for instant verification. Scanning the QR code can directly fetch certificate details and blockchain validation status. This simplifies the verification process for employers and institutions. It also reduces the need for manual input of certificate IDs. QR-based systems improve speed and user experience.

8. AI-Based Fraud Detection

Artificial Intelligence can be integrated to detect fraudulent activities and anomalies. AI models can analyze patterns in certificate usage and identify suspicious behavior. This adds an intelligent layer of security to the system. It helps in preventing misuse and unauthorized modifications. AI can continuously improve system reliability through learning.

CHAPTER-9

CONCLUSION

The proposed system, EduCredChain, successfully demonstrates a secure and efficient approach to digital certificate management using a combination of blockchain technology and cryptographic techniques. By generating and storing certificate hashes on the blockchain while maintaining encrypted data in a database, the system ensures data integrity, authenticity, and resistance to tampering. The implementation of issuance, verification, and revocation mechanisms provides a complete lifecycle solution for managing educational credentials.

The system addresses major limitations of traditional certificate verification methods, such as fraud, manual verification delays, and lack of transparency. Through automation and decentralized validation, it enhances trust among institutions, employers, and users. Additionally, the use of widely available technologies makes the system practical and feasible for real-world deployment.

In conclusion, EduCredChain provides a reliable, scalable, and future-ready solution for secure credential verification. With further enhancements such as smart contracts, mobile applications, and large-scale integration, the system has the potential to significantly transform the way educational and professional credentials are issued and verified.

CHAPTER -10

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APPENDIX – I

A.1.1 Python code to Connect to Blockchain Network

The Python code establishes a connection to the blockchain network (Ethereum via Ganache) using Web3 to enable interaction with smart contracts and transaction processing.

```
def __init__(self):
    ganache_url = "http://127.0.0.1:7545"

    self.CONTRACT_FILE = "Certificate.sol"
    self.SOLC_VERSION = "0.8.17"
    self.w3 = Web3(Web3.HTTPProvider(ganache_url))

    self.abi, self.bytecode = self.compile_contract()
    self.w3 = self.connect_blockchain()
    self.contract = self.deploy_or_load_contract()

    self.sqlecc = SQL_ECC.Sql_db()

def connect_blockchain(self):
    if not self.w3.is_connected():
        print("Ganache not connected.")
        sys.exit()

    self.w3.eth.default_account = self.w3.eth.accounts[0]

    print("Connected to Blockchain")
    print("Chain ID:", self.w3.eth.chain_id)
    print("Active Account:", self.w3.eth.default_account)

    return self.w3
```

A.1.2 Certificate Issue

The Python code handles certificate issuance by processing input data, generating a SHA-256 hash, storing certificate details in the database, and recording the hash on the blockchain.

```
def issue_certificate(self,data):

    now = datetime.now()

    name = data["studentName"]
    course_id = data["course"]
    certificate_no = data["certid"]
    issue_date = now.strftime("%d-%m-%y")
    grade = data["grade"]

    cert_hash = self.generate_certificate_hash(
        name, course_id, certificate_no, issue_date, grade
    )

    self.sqlecc.add_record([certificate_no,name, course_id, issue_date,

grade,cert_hash.hex()])

    try:
        tx_hash = self.contract.functions.issueCertificate(cert_hash).transact(
            {"from": self.w3.eth.default_account, "gas": 200000}
        )

        receipt = self.w3.eth.wait_for_transaction_receipt(tx_hash)

        return{"status":1,
               "Message": "issued",
               "certhash":cert_hash.hex(),
               "blockno": receipt.blockNumber}

    except Exception as exc:
        return{"status":0,
               "Error":"EB01",
               "Message": "BlockChianError : "+ exc}
```

A.1.3 Certificate Verification

The Python code performs certificate verification by retrieving stored data, regenerating the hash, and comparing it with the blockchain record to validate authenticity.

```
def verify_certificate(self, data):

    print(data)
    row =
[data['studentName'],data['course'],data['certid'],data['date'],data['grade']]

    # Recompute hash
    recalculated_hash = self.generate_certificate_hash(
        row[0], row[1], row[2], row[3], row[4]
    )

    recalculated_hash = recalculated_hash.hex()

    if isinstance(recalculated_hash, bytes):

        try:
            text = recalculated_hash.decode("utf-8")
            cert_hash = bytes.fromhex(text)

        except Exception:
            cert_hash = recalculated_hash

    elif isinstance(recalculated_hash, str):
        cert_hash = bytes.fromhex(recalculated_hash)
    else:

        return{"status":0,
               "Error":"EB02",
               "Message": "DB_Decryption_Error"}

    try:
        exists, revoked, issued_at, revoked_at =
self.contract.functions.verifyCertificate(
            cert_hash
        ).call()
```

A.1.4 Certificate Revocation

The Python code handles certificate revocation by updating the certificate status in the database and recording the revocation on the blockchain to prevent further use.

```
def revoke_certificate(self, data):
    row = self.sqlecc.select_record(str(data['certid']))
    print(1)
    if not row:
        return {"status": 0,
                "Error": "EDB01",
                "Message": "Data_not_Found"}

    cert_hash_hex = row[5]
    try:
        print(2)
        cert_hash = bytes.fromhex(cert_hash_hex)
    except Exception:
        return {"status": 0,
                "Error": "EDB03",
                "Message": "Hash_Decode_Error"}

    try:
        print(3)
        tx_hash = self.contract.functions.revokeCertificate(cert_hash).transact(
            {"from": self.w3.eth.default_account, "gas": 200000}
        )
        receipt = self.w3.eth.wait_for_transaction_receipt(tx_hash)

        return {"status": 1,
                "Message": "Revoked",
                "certhash": cert_hash_hex,
                "blockno": receipt.blockNumber}

    except Exception as exc:
        print("\nError:", exc)
        return {"status": 0,
                "Error": "EB03",
                "Message": "BlockChain_Revoke_Error"}
```

A.1.5 Database Operations

The Python code performs database operations by inserting certificate data into the database and retrieving records for verification and management processes.

```
def add_record(self, data):
    cert_no, name, course_id, issue_date, grade, hash_val = data

    name_enc = self.rsa_encrypt(name)
    course_enc = self.rsa_encrypt(course_id)
    grade_enc = self.rsa_encrypt(grade)

    self.cursor.execute(""" INSERT INTO records (Certificate_No, Name,
        Course_ID, Issue_Date, Grade, Hash) VALUES (?, ?, ?, ?, ?, ?) """,
       (cert_no, name_enc, course_enc, issue_date, grade_enc, hash_val))
    self.conn.commit()

def select_record(self, cert_no):
    self.cursor.execute("SELECT * FROM records WHERE Certificate_No = ?",

(cert_no,))
    row = self.cursor.fetchone()
    if row:
       cert_no, name, course_id, issue_date, grade, hash_val = row

       name = self.rsa_decrypt(name)
       course_id = self.rsa_decrypt(course_id)
       grade = self.rsa_decrypt(grade)

       return (cert_no, name, course_id, issue_date, grade, hash_val)

    return None
```

A.1.6 RSA Encryption & Decryption

The Python code implements RSA encryption and decryption to securely protect certificate data and ensure that only authorized parties can access or read the information.

```
def rsa_encrypt(self,data):
    key_path = os.path.join(os.path.dirname(__file__), "key", "public.pem")
    with open(key_path, "rb") as f:
        public_key = serialization.load_pem_public_key(f.read())

    encrypted = public_key.encrypt(
        data.encode(),
        padding.OAEP(
            mgf=padding.MGF1(algorithm=hashes.SHA256()),
            algorithm=hashes.SHA256(),
            label=None
        )
    )

    return base64.b64encode(encrypted).decode()

def rsa_decrypt(self,data):

    key_path = os.path.join(os.path.dirname(__file__), "key", "private.pem")
    with open(key_path, "rb") as f:
        private_key = serialization.load_pem_private_key(
            f.read(),
            password=None
        )

    decrypted = private_key.decrypt(
        base64.b64decode(data),
        padding.OAEP(
            mgf=padding.MGF1(algorithm=hashes.SHA256()),
            algorithm=hashes.SHA256(),
            label=None
        )
    )

    return decrypted.decode()
```

APPENDIX – II

A.II.1 Ganache Local Blockchain Setup Screenshot

The Ganache Local Blockchain Setup Screenshot displays the simulated Ethereum environment used for testing the EduCredChain system. It shows multiple generated accounts along with their respective addresses, private keys, and available balances. The interface also presents details of transactions, blocks, and gas usage, providing a clear view of blockchain activity. This setup enables a controlled environment for deploying smart contracts and recording certificate hashes without using a live network

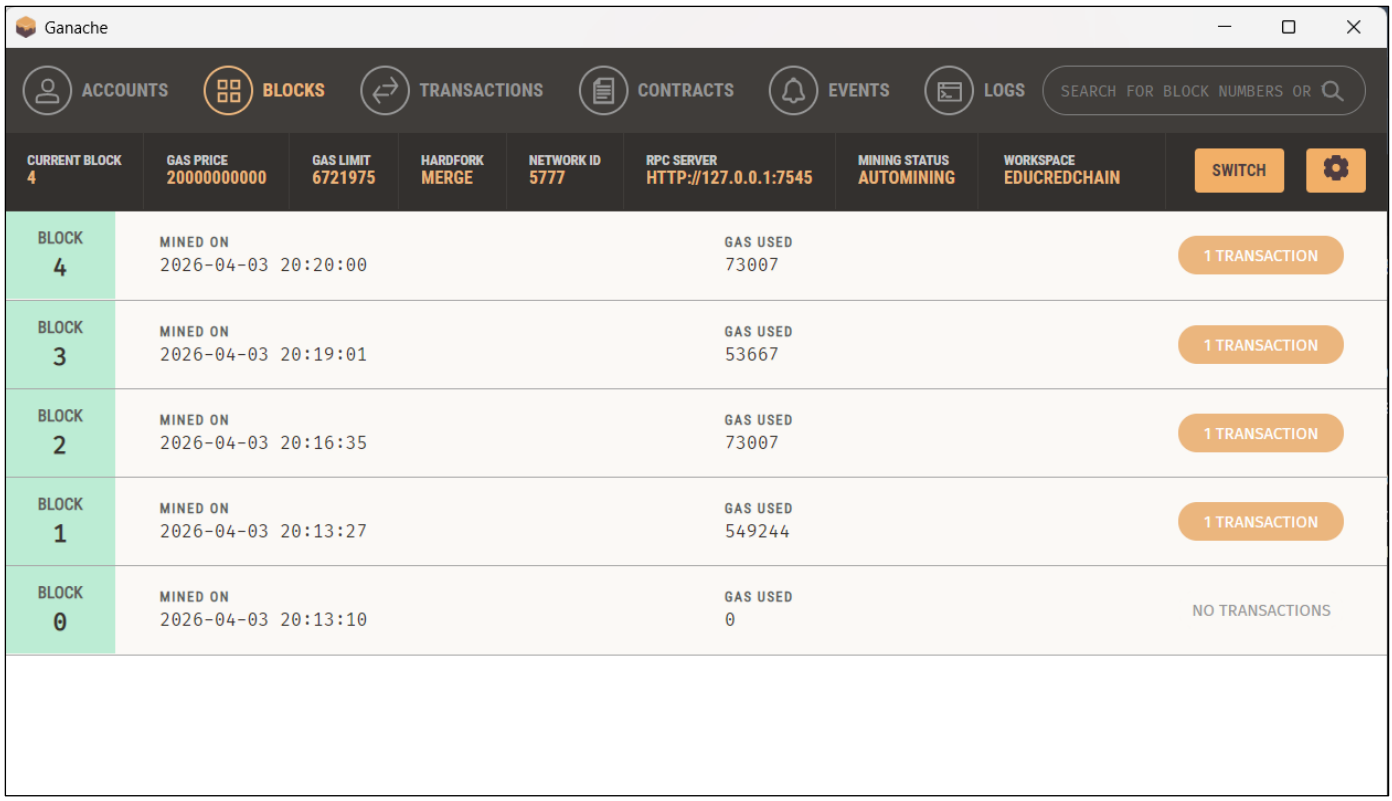

fig A.II.1, Ganache Local Blockchain Setup Screenshot

A.II.2 SQLite Database Structure (certificates.db)

The screenshot displays the structure of the certificates.db SQLite database used in the EduCredChain system. It shows the records table where only the certificate ID is visible in readable form, while all other fields such as name, course ID, issue date, grade, and hash are stored in encrypted format. This ensures confidentiality and protects sensitive certificate data from unauthorized access. The database functions as the off-chain storage layer, supporting secure storage and retrieval during certificate operations.

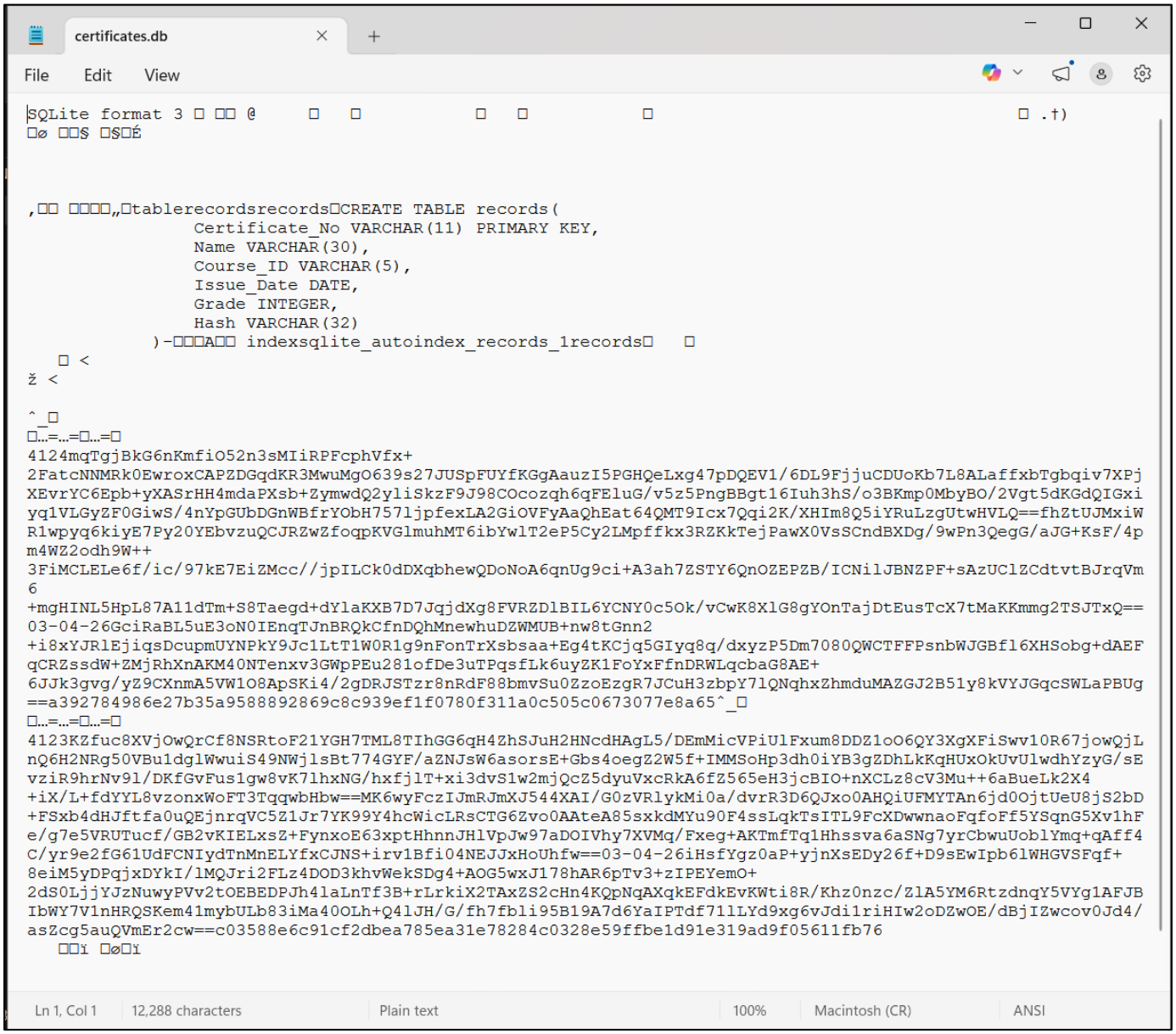

fig A.II.2, SQLite Database Structure (certificates.db)

A.II.3 Flask Backend Execution and Server Logs

The screenshot displays the Flask backend server running the EduCredChain application along with real-time execution logs. It shows the server initialization, API request handling, and responses generated during certificate operations such as issuance, verification, and revocation. The logs provide insight into backend processing, including data handling, hashing, and blockchain interactions. This demonstrates the proper functioning and communication between the frontend, backend, database, and blockchain components.

```
C:\Users\ABHIRAM_S\AppData\Local\Programs\Python\Launcher\py.exe
Connected to Blockchain
Chain ID: 1337
Active Account: 0xFf5f5E9Ec107D555a2Aa90bB412be65A2149Ce59
Loaded Existing Contract
Address: 0x60df339b8bF894f8E3bbE65D81B839a16c151f74
DB in use: [(0, 'main', 'D:\\Mini_Project\\code\\Backend\\database\\certificates.db')]

  EDU CRED CHAIN

←[32mINFO←[0m:     Started server process [←[36m3352←[0m]
←[32mINFO←[0m:     Waiting for application startup.
←[32mINFO←[0m:     Application startup complete.
←[32mINFO←[0m:     Uvicorn running on ←[1mhttp://0.0.0.0:5000←[0m (Press CTRL+C to quit)
←[32mINFO←[0m:     127.0.0.1:64195 - "←[1mOPTIONS /receive HTTP/1.1←[0m" ←[32m200 OK←[0m
←[32mINFO←[0m:     127.0.0.1:64195 - "←[1mPOST /receive HTTP/1.1←[0m" ←[32m200 OK←[0m
= ('4123', 'Rahul', 'Cyber Security', '03-04-26', '85', 'c03588e6c91cf2dbea785ea31e78284c0328e59ffbe1d91e319ad9f05611fb76')
←[32mINFO←[0m:     127.0.0.1:61827 - "←[1mPOST /receive HTTP/1.1←[0m" ←[32m200 OK←[0m
= ('4123', 'Rahul', 'Cyber Security', '03-04-26', '85', 'c03588e6c91cf2dbea785ea31e78284c0328e59ffbe1d91e319ad9f05611fb76')
←[32mINFO←[0m:     127.0.0.1:56536 - "←[1mPOST /receive HTTP/1.1←[0m" ←[32m200 OK←[0m
{'certid': '4123', 'studentName': 'Rahul', 'course': 'Cyber Security', 'grade': '85', 'date': '03-04-26'}


 ******** Active ********


←[32mINFO←[0m:     127.0.0.1:56536 - "←[1mPOST /receive HTTP/1.1←[0m" ←[32m200 OK←[0m
= ('4123', 'Rahul', 'Cyber Security', '03-04-26', '85', 'c03588e6c91cf2dbea785ea31e78284c0328e59ffbe1d91e319ad9f05611fb76')
←[32mINFO←[0m:     127.0.0.1:61323 - "←[1mPOST /receive HTTP/1.1←[0m" ←[32m200 OK←[0m
{'certid': '4123'}
8c6976e5b5410415bde908bd4dee15dfb167a9c873fc4bb8a81f6f2ab448a918 True
1
2
3
←[32mINFO←[0m:     127.0.0.1:61323 - "←[1mPOST /receive HTTP/1.1←[0m" ←[32m200 OK←[0m
←[32mINFO←[0m:     127.0.0.1:61642 - "←[1mPOST /receive HTTP/1.1←[0m" ←[32m200 OK←[0m
= ('4124', 'Rohit', 'Cyber Security', '03-04-26', '85', '43150385239477d6557e52c39bc38dad6b7cc4e92b39ced2034076639d251fda')
←[32mINFO←[0m:     127.0.0.1:57370 - "←[1mPOST /receive HTTP/1.1←[0m" ←[32m200 OK←[0m
{'certid': '4124', 'studentName': 'Rohit', 'course': 'Cyber Security', 'grade': '85', 'date': '03-04-26'}


 ******** Active ********


←[32mINFO←[0m:     127.0.0.1:57370 - "←[1mPOST /receive HTTP/1.1←[0m" ←[32m200 OK←[0m
```

fig. A.II.3 Flask Backend Execution and Server Logs